

Centre for Advanced Studies

Dr. A.P.J. Abdul Kalam Technical University



10-Days Online Short Term course

on

Corrosion and Corrosion Monitoring Techniques: An Academics and Industrial problem

Highlights



Course
Completion
Certificate



Discussion
In Detail



Experts led
Training



10+ hours
Online
training

The examination based on 10 days training will be conducted and passing certificates will be distributed.

Registration Link:

<https://cas.res.in/workshopreg.html>



Coordinator

Dr. Gopal Ji

Assistant Professor

Centre for Advanced Studies

Dr. A.P.J. Abdul Kalam

Technical University, Lucknow

Eligibility: B. Tech (4th Year ME / MME/ AC/CE),
B.Sc/M.Sc chemistry and all interested students

Date : 4-13 April 2022, **Mode:** Online,

Number of Seats: 100

Registration Fee – Free

Limited seats (First Come First Serve)

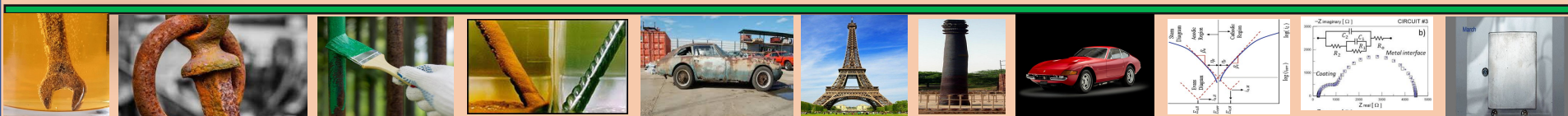
Last Date of Registration: 30 March, 2022

Contents

- ✓ Corrosion Definition and examples, Origin of corrosion, Basics of Corrosion
- ✓ Types of corrosion, detailed discussion about each types Need for corrosion control, Corrosion Control Methods
- ✓ Need and Techniques of Corrosion Monitoring, detailed discussion on Weight loss measurements at room as well as at high temperature, Adsorption isotherms analysis, Gibbes free energy
- ✓ Electrochemical Measurements: Detailed discussion about EIS, Tafel Polarization , Linear Polarization Resistance and OCP

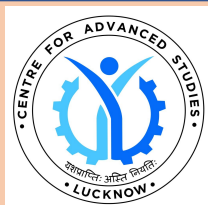
After Completion The Course

- ✓ You will be able to know how to start work in corrosion
- ✓ UG/PG/Ph.D students will be able to understand the basics of corrosion
- ✓ PG and PhD students will be able to write the manuscript on corrosion more effectively and scientifically.



Centre for Advanced Studies, Dr. A.P.J. Abdul Kalam Technical University

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SCHEDULE

Session			
Days	Timing of Class: 4 to 5 PM Daily on Google meet		
Day 1	Introduction to Corrosion: Origin of Corrosion, Reasons for Corrosion, Basic Concepts in Corrosion	Day 7	Electrochemical Corrosion Monitoring : Basics, OCP, Key findings based on OCP, Introduction to Polarization
Day 2	Types of Corrosion : Electrochemical theory of corrosion, Types of corrosion: galvanic, Crevice, Pitting, SCC, Intergranular, Selective Leaching	Day 8	Electrochemical Corrosion Monitoring : Types of Polarization, Over potential, Tafel Polarization curves , Exchange and Corrosion Current Density
Day 3	Types of Corrosion : Erosion Corrosion, Corrosion fatigue, microbial Corrosion, Pipeline Corrosion , Stray Current Corrosion, Need of corrosion prevention	Day 9	Electrochemical Corrosion Monitoring : Mixed potential Theory, Ecorr, Icorr, Tafel Extrapolation, Key findings based on Tafel curves, Solved examples , Impedance, EIS
Day 4	Corrosion Control Methods: Corrosion Control Methods: Material Selection, Design, Coating, Cathodic protection, Anodic Protection, Anodizing	Day 10	Electrochemical Corrosion Monitoring : Nyquist and Bode Curves, Curve Fitting, EEC, Interface analysis, Solved examples
Day 5	Corrosion Control Methods and Corrosion Monitoring: Inhibitors, Need for corrosion monitoring, Monitoring techniques, Hydrogen Evolution Measurements	Examination will be held online through Google forms on Day 12. The examination will be objective type and containing 40 questions. The duration will be 30 minutes. Top 5 candidate will be given examination passing certificate mentioning their position and marks scored. Other will be having examination certificate having marks scored only.	
Day 6	Corrosion Monitoring techniques: Weight loss Measurements at room and High Temperature, Adsorption Isotherm, Arrhenius Plot, Activation Energy, Enthalpy, Key conclusions		

